

MEHUL DADHICH

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Summary

AI/ML Engineer with industry experience shipping production computer-vision systems and building LLM-powered agents and NLP applications. Works across the ML lifecycle, from fine-tuning and evaluation to GPU inference optimisation (CUDA, FP16) and deployment behind FastAPI services, databases and real-time dashboards.

Experience

Infrax.ai

Jan 2026 – Jul 2026

AI Engineer Intern

Remote

- Built the **computer-vision module (TMCS)** of an Advanced Traffic Management System, using **YOLOv8 and YOLO11** to detect road users and flag stopped vehicles, accidents, fire, smoke and fallen objects across **live CCTV feeds**.
- Engineered continuous **RTSP ingestion and inference workflows** for CCTV and IP cameras with Python, OpenCV, MediaMTX and FFmpeg, processing multiple live sources in real time.
- Converted raw detections into structured **incident events** served through **FastAPI** and **PostgreSQL**, powering operator dashboards and real-time alerts.
- Deployed and tuned the pipeline for **continuous 24/7 processing**, addressing frame latency, stream drop-outs, GPU/CPU utilisation and inference throughput.

Projects

RoadGuard – Real-Time Accident Detection | PyTorch, YOLO, ByteTrack, LSTM, Qwen3-VL, OpenCV, FFmpeg

- Designed a multi-stage pipeline: **YOLO detection** → **multi-object tracking** → **30-step LSTM** over trajectory features → **VLM verification**, separating real collisions from normal traffic on video files, webcams and RTSP streams.
- Fine-tuned the vehicle detector to **F1 0.849** on 1,168 held-out frames (IoU 0.5) and ran per-frame detection at **26.9 ms (p50)** on a laptop RTX 3050 with **FP16** inference.
- Cut VLM verification latency **11× (5.5 s → 0.47 s median)** by reformulating the prompt from detailed JSON output to a constrained yes/no query.
- Built a **threaded single-frame buffer** that drops stale frames to keep live latency flat, and an **event state machine** that saves one snapshot and one H.264 clip (≤ 30 s) per incident.

Warehouse Monitoring System | YOLO11, OpenCV, EasyOCR, FastAPI, PostgreSQL, Redis, Docker, React

GitHub

- Built a full-stack warehouse intelligence platform running **YOLO11** on RTSP camera streams to detect, track and count vehicles and people in real time, with **ANPR** (YOLO + EasyOCR) for entry/exit logs.
- Exposed **FastAPI REST** and **WebSocket** streams for live dashboards and alerts; containerised with **Docker Compose** and **Nginx**, with JWT/OAuth2 auth and role-based access.

AI Task Manager – LLM Agent | Python, LangChain, LangGraph, Gemini, Node.js, Firebase, Socket.IO

GitHub

- Built an **agentic LangGraph workflow** that turns natural-language instructions into structured tasks (priority, category, due date), coordinating state and tool execution across steps.
- Integrated **Google Gemini** for classification and date extraction with a keyword fallback for LLM outages; Node.js/Express backend with Firebase Auth/Firestore and **Socket.IO** real-time alarms.

Technical Skills

Languages: Python, SQL, JavaScript, Java

ML & Deep Learning: PyTorch, NumPy, model fine-tuning and evaluation, LSTM, inference optimisation (CUDA, FP16)

Computer Vision: YOLOv8/YOLO11 (Ultralytics), OpenCV, object detection and tracking (ByteTrack), OCR/ANPR, VLMs

Generative AI & NLP: LLMs, AI agents, LangGraph, LangChain, RAG, embeddings, vector DBs, MCP, tool calling, prompt engineering

Video & Streaming: RTSP, FFmpeg, MediaMTX, multithreading

Backend & Tools: FastAPI, REST, WebSockets, PostgreSQL, MySQL, Redis, Firebase, Node.js, React, Docker, Nginx, Git, Linux

Education

VIT-AP University

2022 – 2026

B.Tech. in Computer Science and Engineering (Artificial Intelligence & Machine Learning)

CGPA: 7.67/10

DAV Public School, Ashok Vihar, Delhi

2019 – 2021

Class XII (CBSE): 86% · Class X (CBSE): 90.2%

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** – Oracle
- Hedera Certified Developer Associate (HCDA)** and **Hedera Certified Foundation (HCF)** – Hedera
- Generative AI and ChatGPT – BlackBucks Education · Introduction to Software Engineering – IBM / Coursera